CLIMB at VBS 2027: Multi-Signal Video Retrieval with Collaborative Ad-Hoc Search

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Abstract. We present CLIMB, an interactive video retrieval system built for the Video Browser Showdown 2027. Because a single visual embedding often lacks the precision needed to capture fine-grained cues like signage or spoken dialog, CLIMB retrieves information across four complementary signals: visual embeddings, on-screen text, generated captions, and speech transcripts. By combining rank-based fusion with a hybrid of lexical and semantic matching, the system ensures broad retrieval coverage. Additionally, queries can be chained into temporal sequences, providing increased flexibility while significantly narrowing the results set. For ad-hoc video search, CLIMB treats retrieval as a shared team activity by automatically hiding teammate-submitted shots from all active results. Crucially, core functions such as search, browsing, and submission remain fully operational even if the shared session service becomes unreachable.

Keywords: Video Browser Showdown · Interactive Video Retrieval · Collaborative Search · Semantic Embedding.

1 Introduction

The Video Browser Showdown (VBS) [3, 4, 8] is an interactive video retrieval competition that asks participants to solve known-item search (KIS), ad-hoc video search (AVS), and visual question-answering (VQA/QAS) tasks against large-scale video collections. For VBS 2027, the main collection is V3C [9] with 28,450 videos, roughly 3,800 hours, and about 4.7 TB full-resolution video data, segmented into 4.14 million master shots, along with the smaller collections MVK [13] and GynSurgLHE [6].

Successfully completing these tasks presents two fundamental challenges. A single visual embedding is often too vague to capture precise task hints, such as specific on-screen text or spoken dialogue. And because AVS scoring rewards distinct correct shots, a team of n operators searching independently spends a substantial share of its submissions re-finding what a teammate has already sent. CLIMB addresses both. It indexes every video as scenes and keyframes, retrieves over four independent signals, fuses them by rank, and treats an AVS task as a single shared activity rather than n parallel ones, vanishing duplicate scenes from everybody's list.

Concretely, we contribute:

1. Retrieval at Keyframe-level that integrates four heterogeneous signals by rank (Section 4.1).
2. A temporal query language, `A >> B >>(120) C`, in which every stage is itself a full query (Section 4.3).
3. A collaborative, network-resilient AVS mode (Section 6).
4. An index engineered for everyday hardware: binary-quantized HNSW with oversample-and-rerank (Section 3).

2 System Overview

CLIMB is built from six components (see Fig. 1). Decoding each video, segmenting it into scenes, selecting keyframes, computing embeddings, and text recognition in frames via OCR, as well as captioning and transcribing, happens in an offline pipeline described in Section 3. The resulting scenes, keyframes, and four indexes that the retrievers query land in a PostgreSQL database.
At query time, the system runs as two processes, a Node.js backend transmitting media to the frontend, handling DRES communication and collaboration features [1] as well as a Python $<$ engine handling models, query embedding, and rank fusion. Both connect to the database: the backend reads metadata directly, delegating only search execution to Python. The frontend (Section 5) communicates exclusively with the Node.js backend.
For scenarios where network reliability is a concern, CLIMB supports fully localized deployments, serving media assets directly from localhost. Additionally, the shared and optional collaboration service (Section 6) only transmits a few kilobytes of JSON polled every four seconds.

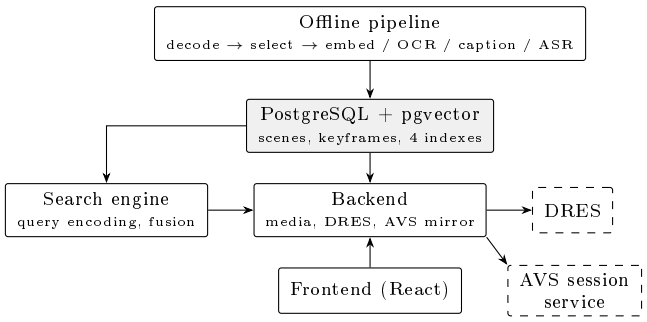


Fig. 1. CLIMB architecture. All components can run locally on each machine, except for two external services: DRES and the AVS session service, which is deployed once.

[1] A local copy of the shared Ad-hoc Video Search state maintained by the AVS Session Service, enabling multiple CLIMB instances operated by different team members to collaboratively search while avoiding duplicate scene submissions.

3 Indexing Pipeline

3.1 Scenes and Keyframe Selection

The first step of the indexing pipeling is shot segmentation and keyframe extraction, which is done with TransNetv2 [11] for V3C and a specific keyframe extraction algorithm [10] for the LHE [6] and MVK videos [12], which is based on ORB keypoint tracking. We start by selecting the most typical frame and then repeatedly add the frame that is most different from the already selected keyframes. This selects a broad coverage of keyframes. Afterward, we reject frames that are nearly flat and have almost no edges. This reduces the number of keyframes sampled from our subset (Section 7).

3.2 Feature Extraction

CLIMB extracts four signals per scene, ensuring a diverse set of retrieval possibilities. They are deliberately not applied equally. Two keyframe- and two scene-based signals. For OCR we leverage the PP-OCR [2] model, while visual embeddings with 1024 dimensions are extracted with SigLIP2 [14].
Given the high computational cost and intra-scene visual redundancy, captioning is restricted to each scene's first keyframe. For full-scale captioning, we apply Qwen2-VL-2B-Instruct [16]. Audio transcription is handled by Whisper [7].
In order to encode text into semantic embeddings, we use multilingual-e5-base [15], which performs encoding with 768 dimensions. For efficient retrieval, all generated embeddings are stored in an HNSW index [5] over binary-quantized vectors. To maintain search accuracy while minimizing memory overhead, initial search candidates are reranked against their full-precision vectors. This quantization significantly reduces the memory footprint, requiring only 5.3 GB for 12.4M keyframes compared to 28 GB for half-precision storage.

4 Retrieval Model

4.1 Rank-Based Fusion at Keyframe Level

As discussed above, we use four different encoders to encode as broad a spectrum of information as possible. In order to avoid sensitive and model-specific calibration, we use a ranked-based fusion approach, explained below.
Each signal returns a list of ranked objects. While the visual and OCR signal natively return a list of ranked keyframes, the caption and audio signals are scene-based. Thus, each keyframe inherits the rank of its parent scene. The keyframes are then merged via weighted Reciprocal Rank Fusion (RRF) [1]. This works similarly to normal RRF but weighs the signals depending on their source, allowing us to highlight keyframes found via specific encoders, i.e., OCR, where a lexical hit is much more likely to belong to the correct scene than a similar

visual embedding. As by [1], the fused ranking score for a keyframe $d \in D$ is defined as:

$$RRFscore(d \in D) = \sum_{r \in R} \frac{w(r)}{60 + r(d)} \tag{1}$$

Where $r(d)$ is the rank assigned by encoder $r \in R$ with weight $w(r)$. R denotes the set of information modalities. The constant 60 reduces the dominance of top-ranked keyframes within individual information modalities, allowing evidence from multiple sources to contribute to the final score. This constant was empirically determined in [1].

4.2 Semantic and Lexical matching

Although cross-modal visual retrieval with SigLIP2 is inherently semantic, the remaining text-to-text signals could theoretically rely on lexical matching. However, strict keyword matching on generated captions suffers from severe vocabulary mismatch. For instance, a query for "law enforcement" shares zero lexical overlap with a highly relevant caption describing "a police uniform" In fact, the terms "law" and "enforcement" never appear within our extracted captions.
While ASR transcripts are embedded semantically for conceptual and cross-lingual search, users can invoke a *said:* operator for exact lexical matching of specific quotes. In contrast, OCR operates as a purely lexical signal to pinpoint exact on-screen text provided in task hints. Semantic matching here is redundant, as the visual embedding already captures general concepts.

4.3 Query Language and Temporal Chains

Some retrieval tasks require describing not a single scene, but a temporally ordered sequence of scenes, such as a bridge followed by a beach. A plain text query searches all four signals at once, and the operator prefixes in Table 1 restrict it to a single source, whilst `>>` joins queries into ordered temporal chains. The chaining makes the prefixes much more powerful. A query `A >> B >> C` has three stages, each of which is a complete query on its own. `text: pegasus >> mountain bike race` finds a scene with a text that reads "pegasus", followed by a scene that matches a mountain bike race, independent of a single encoder. An unannotated temporal chain searches for scenes within the same video, where consecutive scenes are at most 60 seconds apart. The maximum temporal distance can be specified using an annotation such as `(120)` or `(d120)`, allowing a gap of up to 120 seconds. This distance can be increased to a maximum of ten minutes, with each chain containing up to four stages.

Table 1. Query syntax. Prefixes restrict searches to specific signals, while >> chains queries temporally within a single video.

Input	Effect
`a drawing of people on a beach`	Global search (all signals)
`text: state capitol`	On-screen text only
`said: new york`	Spoken audio only
`-video:00191`	Exclude a video (repeatable)
`a video game sunset >> door`	Sequential search (same video)

5 Interactive Search Interface

The CLIMB user interface is shown in Figure 2. A search typically follows an iterative interaction loop. The operator enters a query that returns a grid of keyframes. Selecting one displays a five-second scene preview starting at the corresponding timestamp. The filmstrip located under the video player provides an overview of all the keyframes, allowing efficient navigation between them. Keyframe times are automatically populated but remain editable, to fine-tune the target submission frame. In addition, there are a variety of buttons for quick search refinement. To keep this cycle responsive, query results are cached and paginated: the initial query executes the search, while subsequent pages are loaded from cache upon scrolling. *Submit to DRES* sends the selected item to the evaluation server. *Exclude video from search* removes a video from the results, and when textual queries fall short, *Find Similar* enables visual query-by-example. Finally, dedicated UI inputs support VQA tasks, allowing operators to submit text-only or combined text-and-shot answers.

6 Collaborative Ad-Hoc Video Search

Since AVS scoring rewards distinct correct shots, independently team members tend to identify the same obvious results first, resulting in a large fraction of duplicate submissions. We address this problem by making an AVS session a shared state. Submitted scenes are consequently hidden from the results of all session participants. Other scenes in these videos get marked as covered but not hidden to promote a diverse submission set, while allowing for the submission of multiple distinct scenes from the same video.
Importantly, searches and submissions do not depend on the speed or availability of the AVS collaboration service. To ensure this separation, CLIMB sends the submission request to the DRES server before attempting to communicate with the independently deployed collaboration service.

7 Preliminary Results

While implemented for the full V3C collection, our reported metrics were measured on a 200-video subset (25 h, 14,345 shots, 35,558 keyframes).

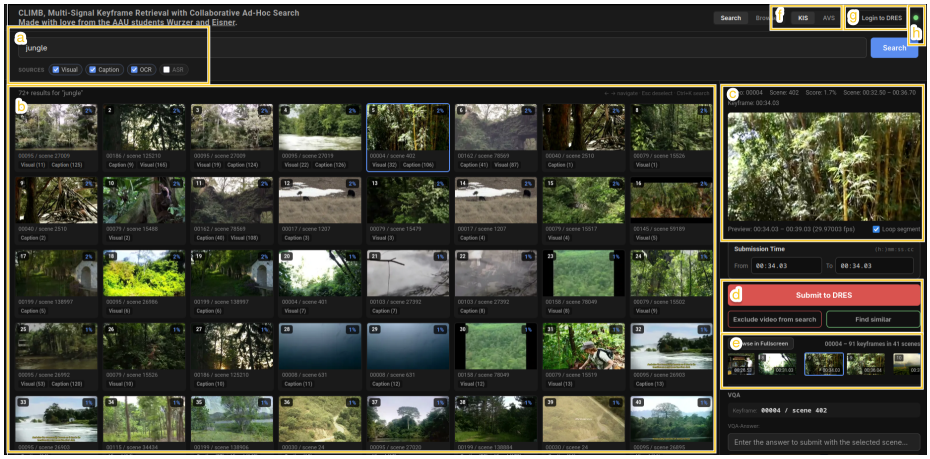


Fig. 2. The CLIMB interface: (a) query bar and source filters, (b) scene result grid, (c) detail panel and video player, (d) submission and search-refinement controls, (e) keyframe filmstrip, (f) AVS session tab, (g) DRES connection controls, and (h) system health status.

Compared with exhaustive exact search (108 ms), oversampling 1,000 candidates before exact reranking recovers 15 of the true top 20 in 29 ms, 2,000 candidates recover 18 in 54 ms, and 4,000 all 20 in 91 ms. The remaining discrepancy is due to the binary quantization rather than the index: the ANN stage ranks by Hamming distance over 1-bit vectors. CLIMB defaults to 1,000 candidates, while keeping this parameter configurable.

Although a far simpler CLIMB prototype won our university's course-internal retrieval competition, VBS presents a much greater challenge. Our latency measurements are limited to a small subset rather than the full-scale collection, and neither MVK nor GynSurg resembles the imagery SigLIP2 was trained on.

8 Conclusion

In this paper, we present CLIMB, an interactive video retrieval system for VBS 2027. It uses rank-based fusion to combine visual embeddings, on-screen text, generated captions, and transcripts at the keyframe level while supporting temporally chained queries. A collaborative mode hides shots a teammate already submitted from everyone's results, while search, browsing, and submission remain unaffected when the shared session service is unavailable.

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